數位訊號處理原理與應用

Principles and Applications of digital Signal

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Video Signal Processing and Application Lab.

Text & Reference books

- Text book：
  - Discrete-Time Signal Processing (3rd ed.), A. V. Oppenheim and R. W. Schafer, Prentice Hall, 2009.(全華代理) (請使用合法教科書)
- Reference books：
  - 1. Digital Signal Processing, A Computer-based Approach 4th Edition, Sanjit K. Mitra, Mac Graw Hill, 2009
  - 2. "Signals and Systems", 2nd Edition, 2002, by Simon Haykin and Barry Van Veen, Wiley(歐亞書局代理)
  - 3. 教育部資通計畫教材

Contents

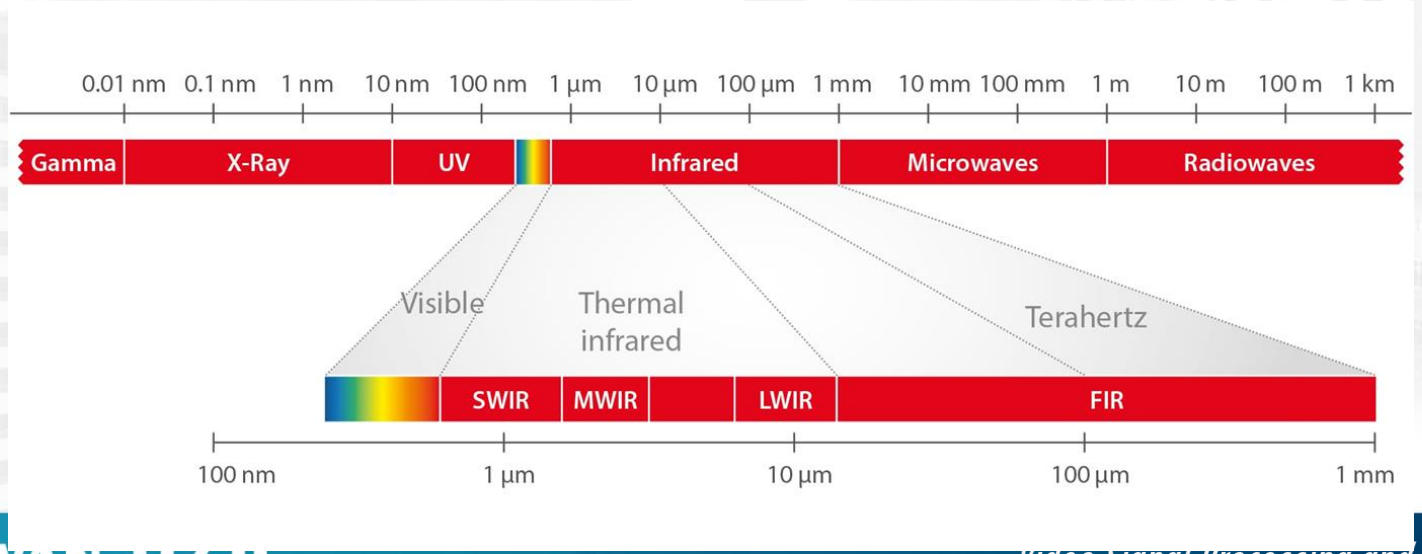
- Part I: Review of Signals and Systems (Chapter 2-3)
  - DSP Systems and Applications
  - Introduction to Discrete-Time Signals and Processing
  - The Z-transform
- Part II: Transform Methods for DSP (Chapter 4-5,8-9)
  - Sampling Theory
  - Examples of Digital Filters
  - The Discrete Fourier Transform (DFT)
  - Fast Fourier Transform
- Part III: Digital Filter Design (Text Chapter 6-7)
  - Type of Filters --- FIR and IIR
  - Filter Design Techniques

Chap1

Introduction to Signals and Signal Processing

Signals

- Speech(e.g. sampling frequency 8KHz:64Kbps → 2.4Kbps),Biomedical signals, Audio(e.g., MP3, sampling frequency 44.1KHz) and Sound, Video and Image, Radar signals…



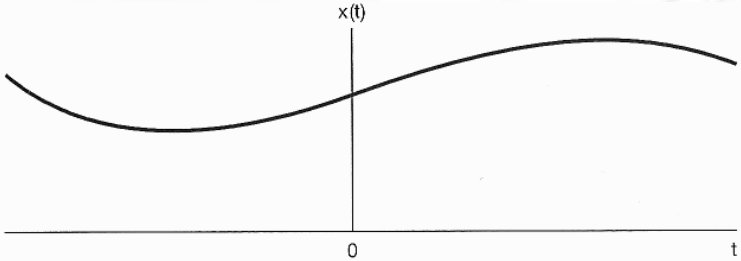
Continuous-Time & Discrete-Time Signals

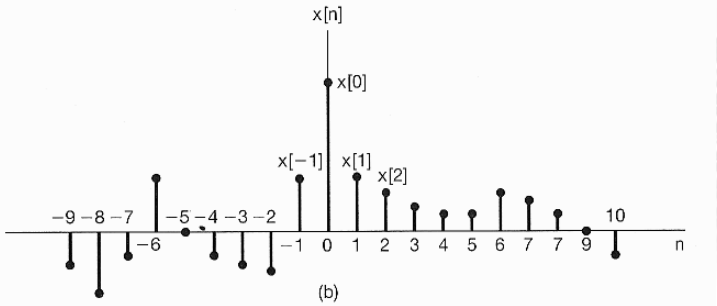
Key concepts (i) continuous and discrete signals; (ii) energy; (iii) average power

Examples and Mathematical Representation

signals

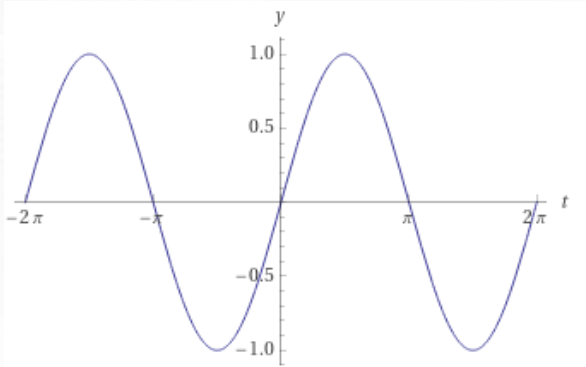
- Continuous-time (CT) signals
- Discrete-time (DT) signals





For a continuous-time (CT) signal, the independent variable is always enclosed by a parenthesis (·)

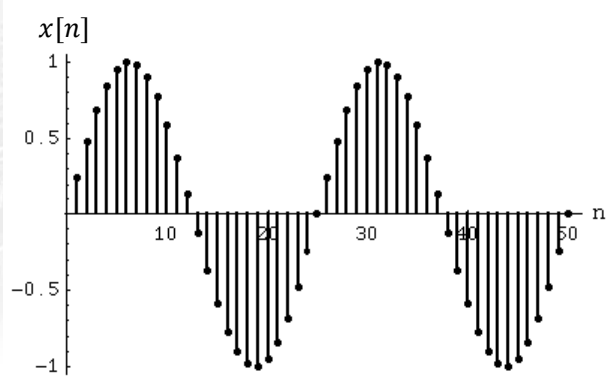
Example: $x(t)$, $y(t)$, $z(t)$, $A(x, y)$, …….



$$y(t) = \sin(t)$$

For a discrete-time (DT) signal, the independent variable is always enclosed by a brackets [·]

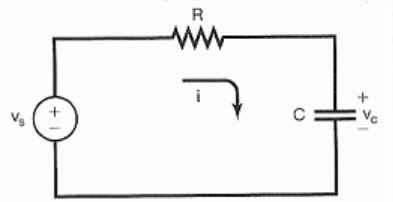
Example: $x[n]$, $y[n]$, $z[n]$, $A[m, n]$, …….



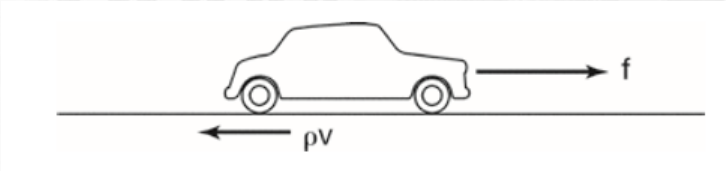
$$x[n] = \sin[2\pi f \cdot n]$$

- Examples of continuous signals

(1) Circuits (voltage, current, electric charge)



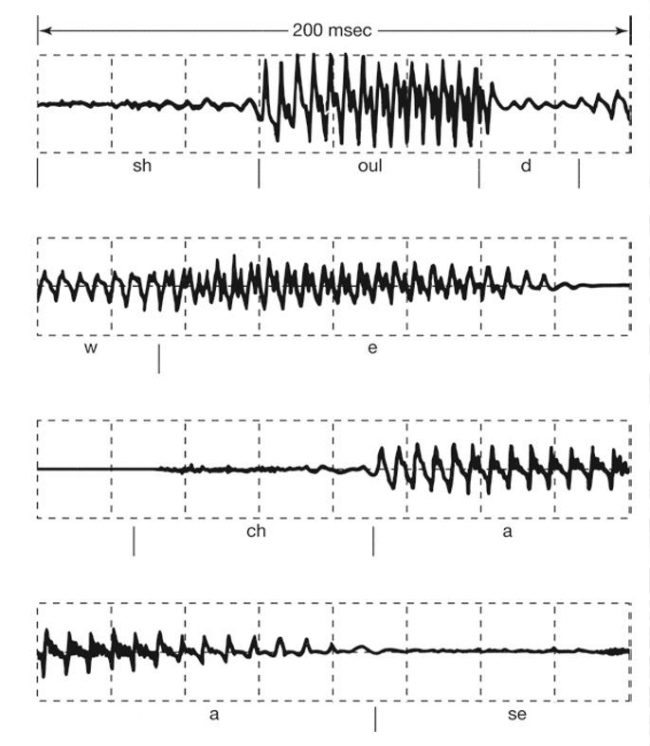
(2) Motion (location, velocity, acceleration)



- Examples of continuous signals

(3) Speech (acoustic pressure)

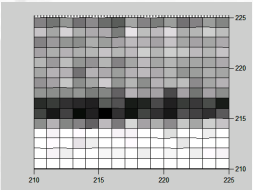
Vowel: larger power
Consonant: smaller power

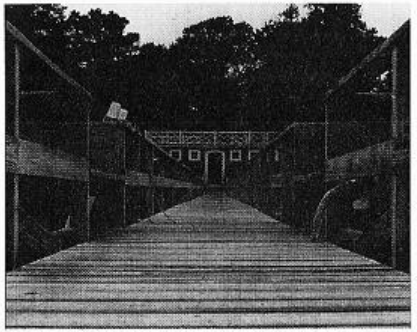


(4) Others (temperature, pressure, light, ECG, …….)

- Examples of discrete signals

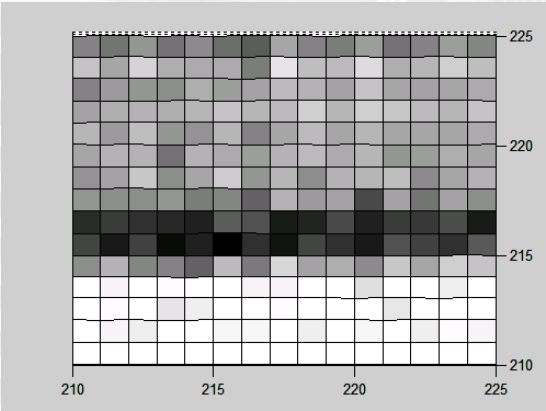
(1) A monochromatic picture





white $\rightarrow$ 255

black $\rightarrow$ 0

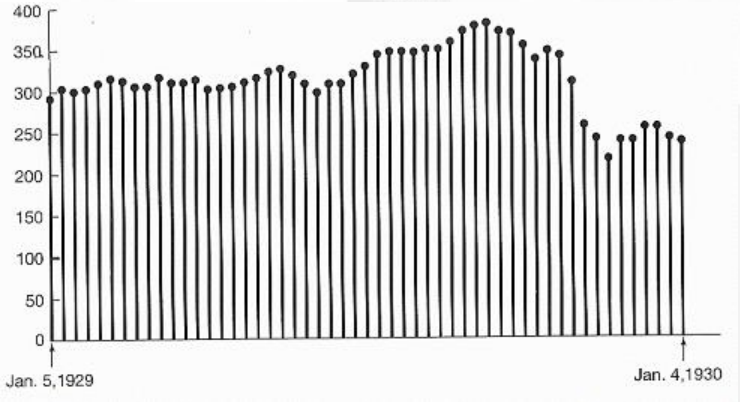


117	151	112	136	110	95	166	131	125	159	114	130	159	133
164	209	173	171	168	126	225	189	184	219	172	180	206	188
154	151	143	174	156	160	186	179	163	193	167	161	167	168
173	179	172	182	195	181	187	205	183	205	196	186	180	193
155	188	149	145	181	131	166	185	166	181	164	164	166	171
176	176	113	178	169	159	177	184	171	182	148	159	174	164
162	198	141	164	202	149	181	142	176	182	189	136	165	176
143	143	148	127	132	97	177	152	160	74	163	119	162	143
62	51	40	32	95	82	28	39	75	35	60	58	77	28
26	67	14	34	3	49	22	69	48	26	81	67	49	91
176	134	129	97	185	120	212	160	173	139	197	166	207	192
243	255	250	255	254	242	242	255	254	222	255	253	239	252
248	255	226	238	255	255	249	255	255	255	230	252	255	255
243	238	255	253	245	247	255	238	255	244	255	237	249	241

- Examples of discrete signals

(2) Stock market index

The weekly Dow-Jones stock market index



Digital Signal Processing

- Digital Signal Processing
  - Sample an analog signals → Signals in digital form.
  - Process by digital processor.
- Objectives
  - Remove interference or noise from a signal.
  - Obtain the spectrum of data.
  - Transform the signal to a suitable form. (e.g. Fourier Transform, Wavelet Transform, Discrete Cosine Transform ...)

DSP

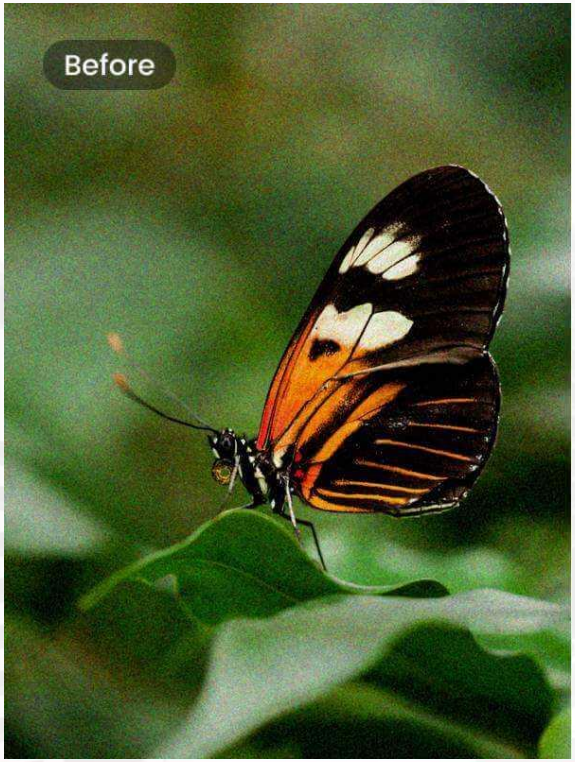
- Advantages
  - Guaranteed Accuracy: depend on number of bits need.
  - Perfect Reproducibility: e.g. digital recording.
  - Advantages is always taken of the advances in semiconductor technology.
  - Flexibility: reprogrammable.
- Disadvantages
  - Speed & cost: fast ADC/DAC are expensive. Bandwidth of 100MHz increased are processed by analog signal method.
  - Design time: DSP hardware increasingly complicated.
  - Finite length effects: limited-bit number of DSP processor.

DSP Applications

- Image/Video Processing
  - Pattern recognition, robot vision, facsimile, animation, data compression…
- Controls
  - Spectrum analysis, position and velocity control of motors, noise reduction…
- Speech/Audio
  - Speech recognition, speech/audio synthesis, text to speech, digital audio, quantization…
- Tele-communications
  - Echo cancellation, adaptive equalization, video conference…
- Vehicle Electronics
  - Telematics, Navigation, Car Safety…
- Biomedical signal processing
  - Patient monitoring, EEG , EKG analysis, Biomedical image processing…

DSP Applications

- Noise removal from image



DSP Applications

- Noise removal from image

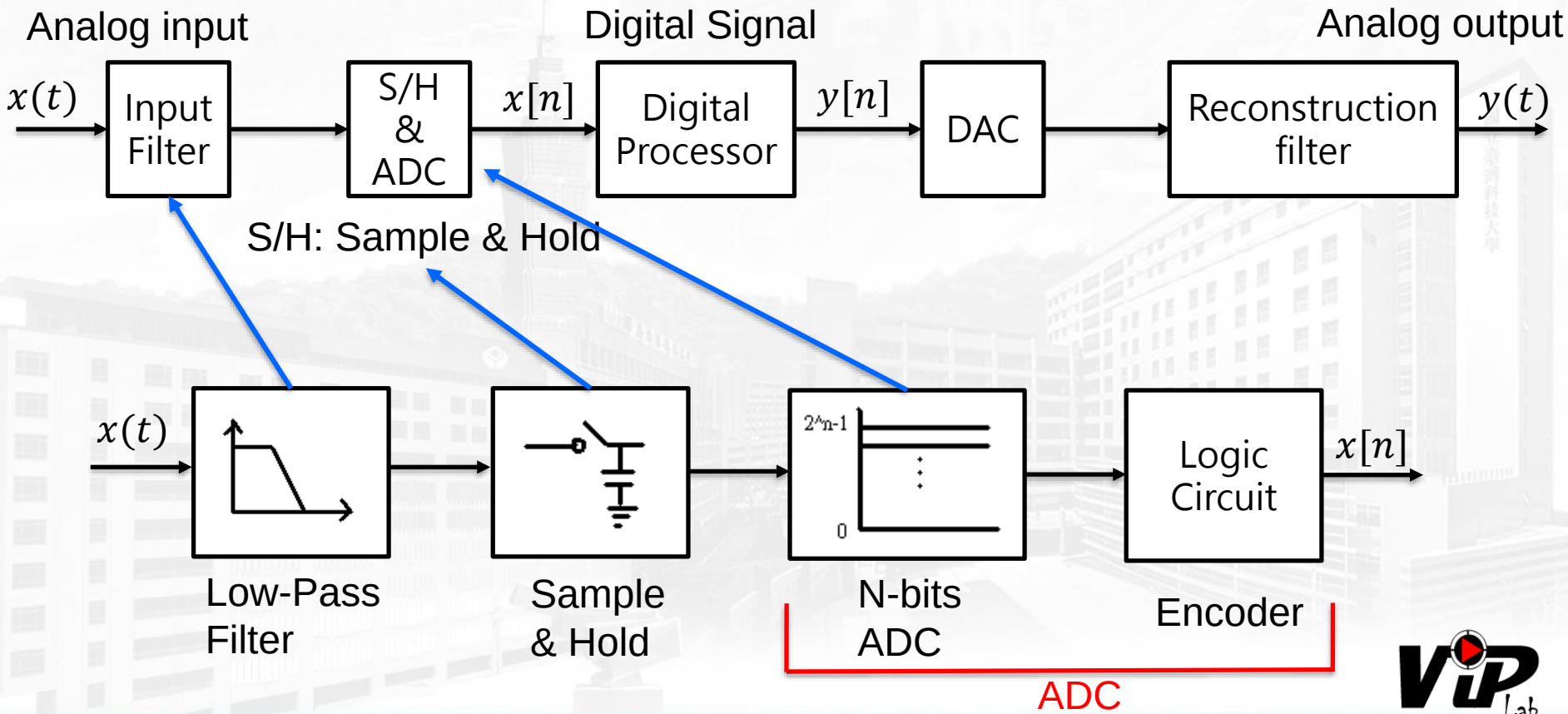
DSP Applications

- Image enhancement

DSP Applications

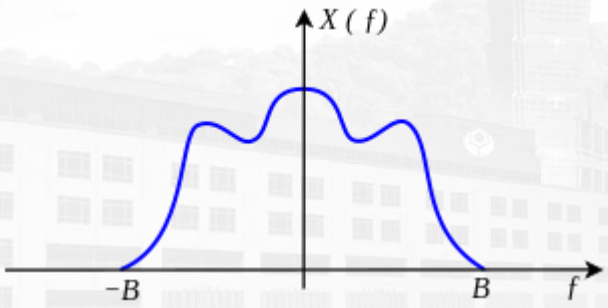
- Key DSP operations
  - Convolution
  - Correlation
  - Filtering
  - Discrete Transforms
    - ✓ DTFT、DFT、FFT、DCT、DWT、Z-Transform
  - Modulation

Real-Time signal processing flowchart

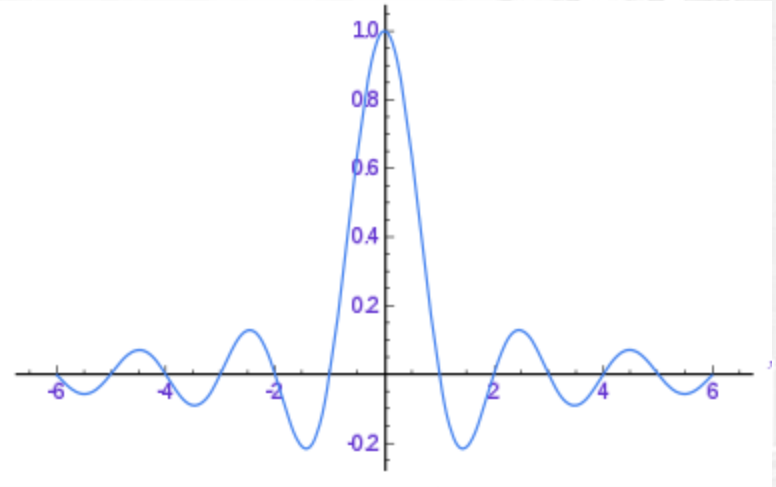


Sampling

- The main function of the low-pass filter is to band-limit the input signal to the folding frequency without distortion

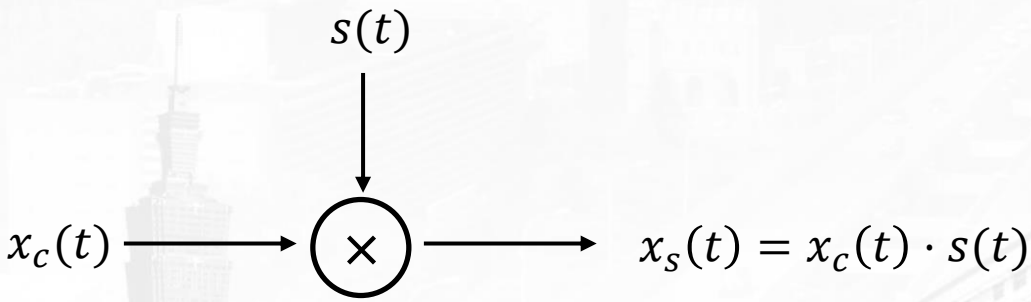


Band-limited signal

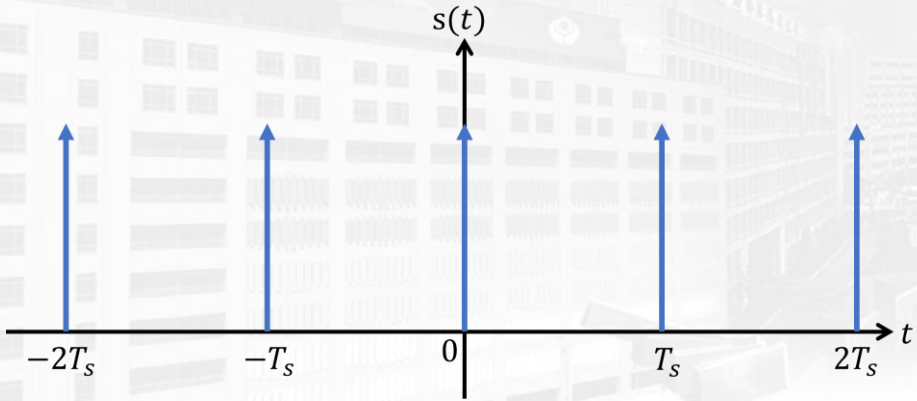


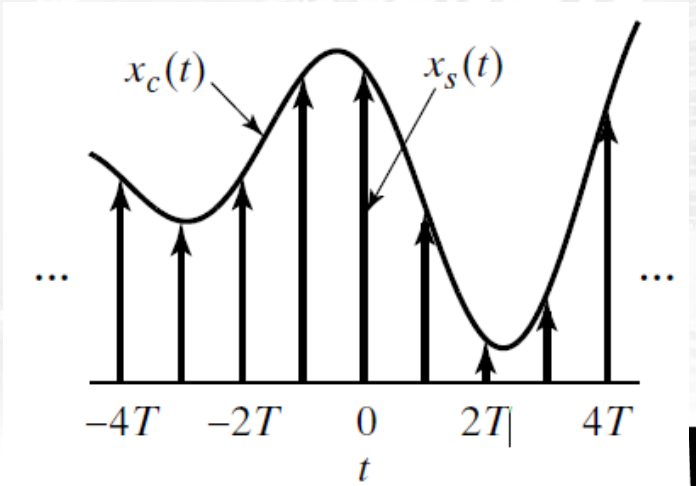
Band-unlimited signal

Sampling

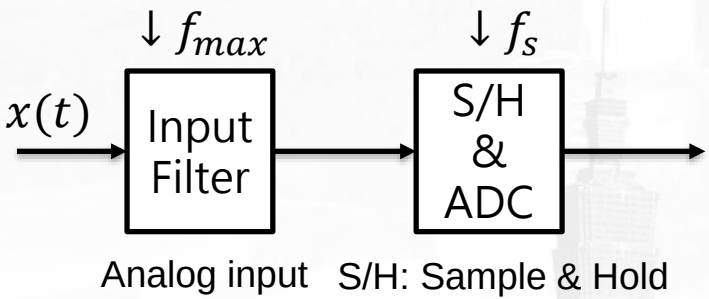


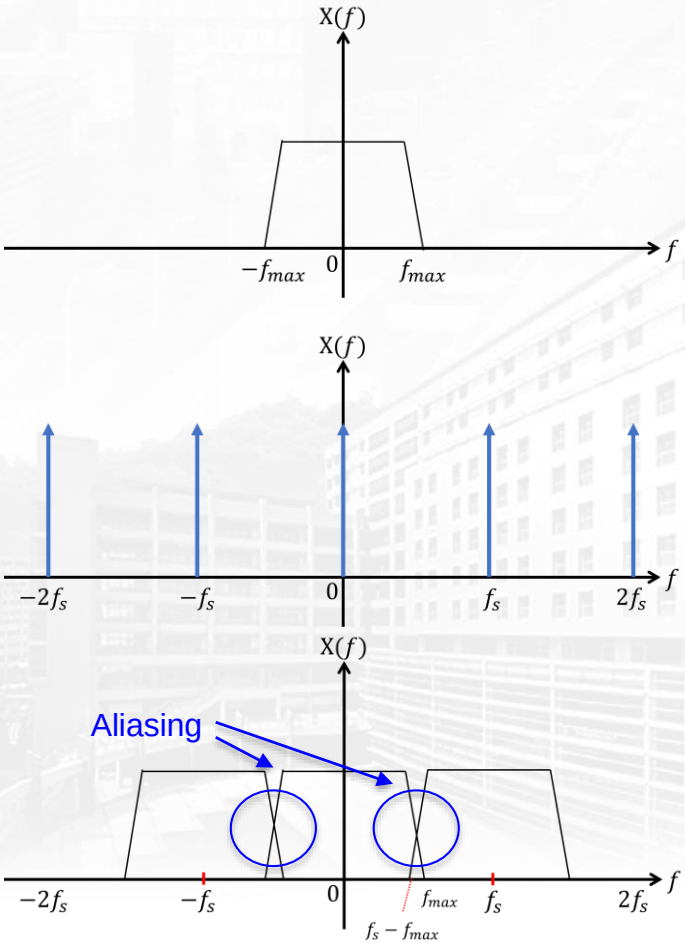
$s(t)$: Impulse train





Sampling problem





$$f_s - f_{max} < f_{max} \Rightarrow f_s < 2f_{max}$$

$\Rightarrow$ Aliasing

Solution

- $f_s \geq 2f_{max}$

  f_s: Sampling frequency
  f_{max}:Maximum frequency of the input signal

- Sampling theorem: $f_s = 2f_{max}$

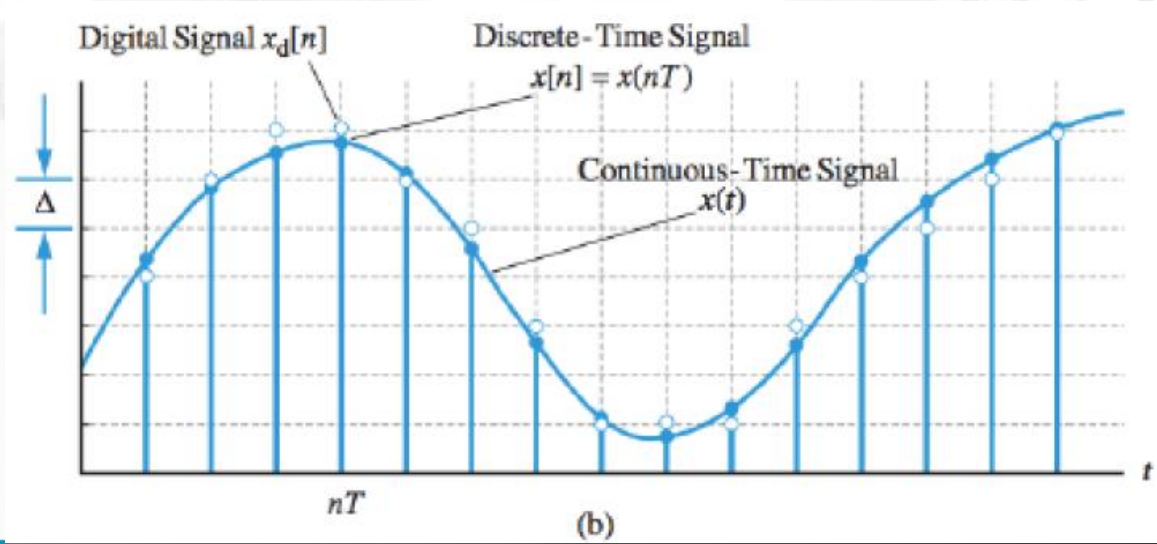
- $\therefore$ Sampling rate $\geq$ Nyquist frequency

  $\Rightarrow$ No distortion (ideally)

Analog-to-digital converter (ADC)

- Quantization converts a continuous-amplitude signal $x(t)$ into a discrete-amplitude signal $x_d[n]$
- In theory, we are dealing with discrete-time signals; in practice, we are dealing with digital signals

Δ:Quantization error



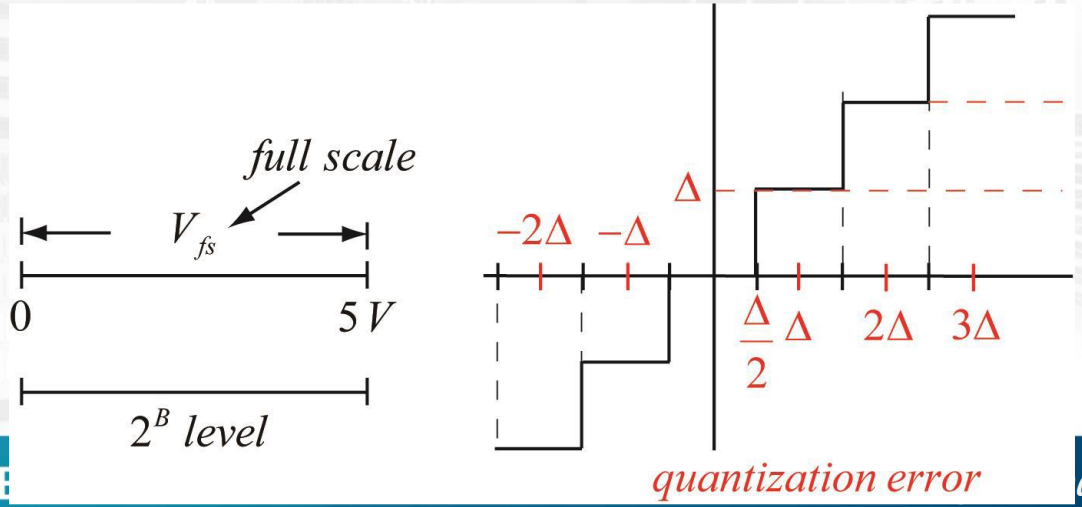
Analog-to-digital converter (ADC)

- Quantization error

  For example : ADC0804 chip, 0~5 Volt, 8 bits

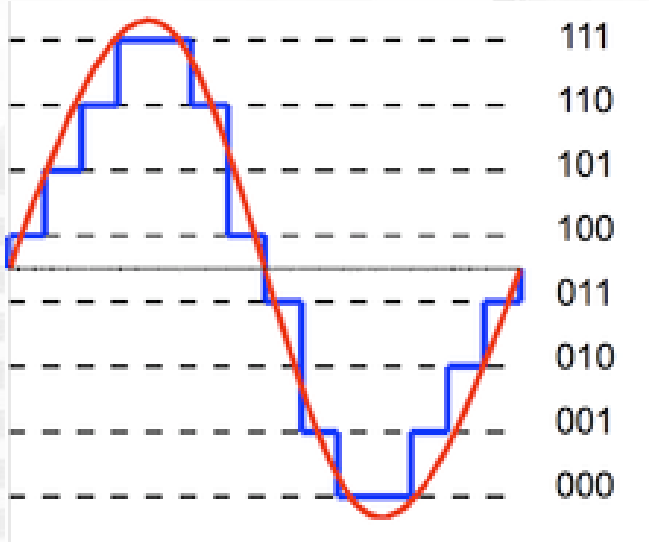
  $\Delta \triangleq LSB = \frac{V_{fs}}{2^B}$ (A B-bit quantizer can represent 2^B different numbers)

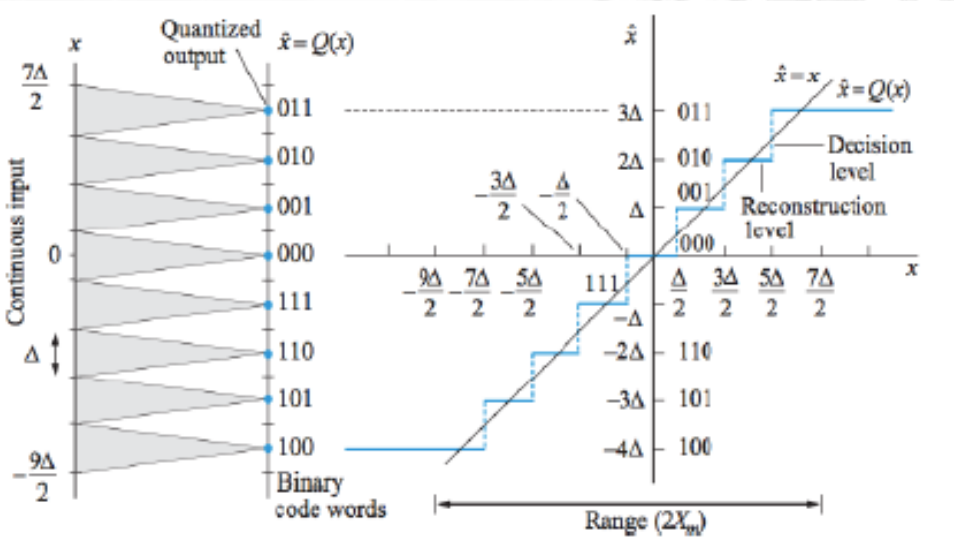
  Range of quantization error $= \pm\frac{1}{2}\,LSB = \frac{1}{2}\frac{V_{fs}}{2^B} = \frac{5V}{2^9}$



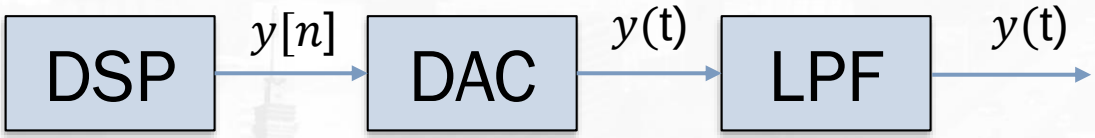
Encoder

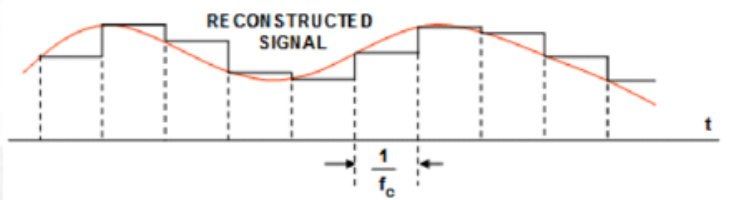
- A B-bit quantizer can represent 2^B different numbers.



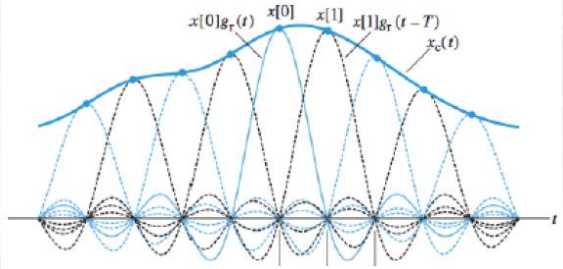


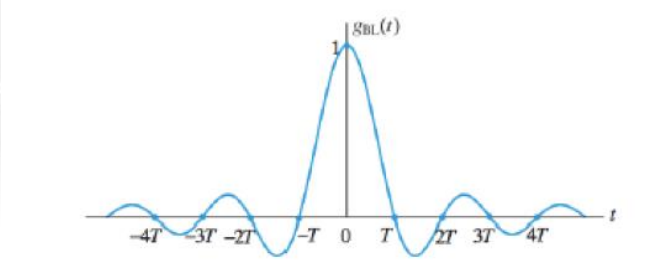
Digital-to-Analog converter (DAC)





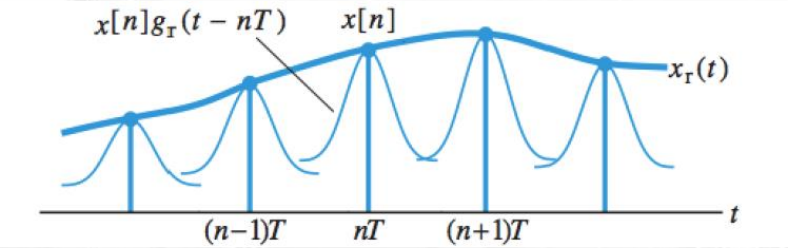
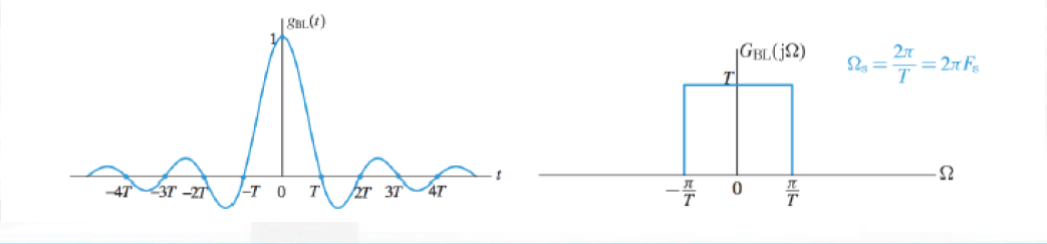
- We note that $g_{BL}(t) = 0$ at all $t = nT$, except at $t = 0$ where $g_{BL}(t) = 1$. Thus, it is always true that $x_r(nT) = x_c(nT)$ regardless of whether aliasing occurred during sampling.





Digital-to-Analog converter (DAC)

- The process of fitting a continuous function to a set of samples is known as an interpolation.



- To understand the meaning and implications of the ideal interpolation we look more closely at the *sinc* function $g_{BL}(t)$.